

Design and characterization of borosilicate glass modified with TiO₂ for enhanced optical and radiation shielding applications

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ABSTRACT

This work investigates the potential of borosilicate glass doped with titanium dioxide (TiO₂) for radiation shielding applications. The study explored the impact of varying TiO₂ concentrations on the glass structure, optical properties, and radiation shielding effectiveness. X-ray diffraction (XRD) and Raman spectroscopy confirmed the amorphous nature of the glasses. The addition of TiO₂ affected the glass network structure, as evidenced by changes in the Raman spectra and density measurements. UV-Vis spectroscopy revealed a red shift in the absorption edge (416–481 nm) and a decrease in the indirect optical band gap (2.91–2.70 eV) with increasing TiO₂ content. The refractive index also increased from 2.54 to 3.54 with TiO₂ concentration. The mass attenuation coefficient (MAC) and linear attenuation coefficient (LAC) increased significantly with TiO₂ addition, indicating enhanced radiation attenuation capability. The inclusion of TiO₂ reduced the half-value layer (HVL), tenth-value layer (TVL), and mean free path (MFP), while increasing the effective atomic number (Z_{eff}). The effective electron density (N_{eff}) and effective conductivity (C_{eff}) showed a slight increase with TiO₂ content. Energy build-up factors (EBF and EABF) indicated a higher rate of radiation intensity increase and energy absorption within the glass due to the presence of TiO₂. Finally, the incorporation of TiO₂ into borosilicate glass offered a promising approach to developing improved optical and radiation shielding glasses.

1. Introduction

Radiation shielding materials are a vital safety measure across various industries. They protect people and the environment from the harmful effects of ionizing radiation (Paul et al., 2024; Moradi et al., 2024). The invisible threat of radiation necessitates robust shielding solutions. Borosilicate glass, a seemingly ordinary material, emerges as a surprising contender in this realm. This essay explores the radiation shielding properties of borosilicate glass matrices, delving into their inherent strengths and potential for improvement. Borosilicate glass offers several advantages as a radiation shielding material. It is relatively inexpensive to produce, exhibits good chemical durability, and can be formed into various shapes. Additionally, its transparency allows for visual inspection and potential integration with optical components (Sayyed et al., 2024a; Hussein et al., 2024). Borosilicate glass, known for

its heat resistance and clarity, boasts a unique internal structure. It comprises a network of silicon-oxygen (Si–O) and boron-oxygen (B–O) bonds, with additional elements like sodium and aluminum often present. These strong covalent bonds create a stable structure, but the key to its radiation shielding lies in its elemental composition (Aloraini et al., 2024; Almousa et al., 2024a).

Radiation interacts with matter through various mechanisms, with photons (gamma rays and X-rays) being a primary concern. Borosilicate glass offers two main modes of defense against these photons: absorption and scattering. Absorption occurs when the photon interacts with an electron within the glass, transferring its energy and effectively disappearing. Scattering deflects the photon's path, reducing its intensity as it travels through the material (Apte and Bhide, 2024). The effectiveness of borosilicate glass in this role hinges on two key factors: density and atomic number. Denser materials pack more atoms per unit

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volume, increasing the probability of photon interaction. Furthermore, elements with higher atomic numbers possess more electrons, making them more proficient at absorbing photons. While the base components of borosilicate glass (silicon and boron) have moderate atomic numbers, their combination offers a reasonable shielding capability (Alzahrani et al., 2024).

However, researchers are constantly striving to improve the performance of borosilicate glass for radiation shielding. One approach involves incorporating elements with high atomic numbers, such as barium or gadolinium. These “dopants” disrupt the glass network slightly but significantly enhance its ability to absorb photons. Borosilicate glass demonstrates promising radiation shielding capabilities due to its composition and structure. Studies have shown that incorporating elements like BaO into borosilicate glasses enhances their photon attenuation properties, with higher mass attenuation coefficients and effective atomic numbers compared to other compositions (Kilicoglu et al., 2023). Additionally, borate glasses containing zinc, bismuth, and lithium have been found to exhibit excellent shielding against gamma-rays, fast-neutrons, and charged particles, with specific compositions showing superior performance in nuclear security applications (Al-Buriah, 2023). Furthermore, the incorporation of silica from rice straw ash into glass systems has shown improved gamma-ray shielding properties, with higher mass attenuation coefficients and effective atomic numbers, making them superior to conventional shielding materials like concrete and commercial glass (Abdelgawad et al., 2023). Additionally, the incorporation of Sb₂O₃ into soda-lime glasses enhances their gamma-ray and neutron shielding parameters, making them effective in radiation safety applications (Sayyed et al., 2020). Furthermore, the ionizing radiation attenuation parameters of glasses like Li₂O–SiO₂ and Li₂O–SiO₂–GeO₂ doped with Y₂O₃ and In₂O₃ have been analyzed, indicating their potential as shields against gamma-rays, charged particles, and neutrons in nuclear radiation scenarios (Alharshan et al., 2023). The addition of Fe₂O₃ to glasses significantly impacts their structural network, enhancing elastic moduli and improving shielding parameters against gamma photons (Almuqrin et al., 2021). Titanium dioxide (TiO₂) significantly enhances the optical and radiation shielding performance of various glass matrices. Research indicates that increasing TiO₂ concentrations in magnesium borate glasses leads to improved radiation shielding, as evidenced by a rising linear attenuation coefficient (μ) and decreasing half-value layer (HVL) values, which reflect enhanced photon attenuation capabilities [15@41]. Similarly, in BaO–TeO₂ glass ceramics, TiO₂ contributes to notable optical properties and varying radiation shielding effectiveness, although it was not the top performer compared to other modifiers like Bi₂O₃ (Mhareb et al., 2024). In zinc barium-borate glasses, TiO₂-doped samples exhibited superior mechanical properties, yet their gamma-ray shielding was less effective than CuO-doped glasses (Alawaideh et al., 2024). Furthermore, TiO₂ demonstrated lower mass attenuation coefficients compared to other oxides, indicating that while it improves certain properties, it may not always be the optimal choice for radiation shielding (Alasali et al., 2024). Another study demonstrated the potential of TiO₂-doped CeO₂–PbO–B₂O₃ glasses as multifunctional materials (Singh et al., 2021). The incorporation of TiO₂ significantly influences the glass structure, physical properties, optical behaviour, and radiation shielding performance. This is evident from the improved linear attenuation coefficient and reduced half-value layer. The partial replacement of SiO₂ with TiO₂ in the Na₂O–CaO–SiO₂ glass system leads to a material with improved mechanical properties and superior radiation shielding performance, particularly in the low-energy range (Albarzan et al., 2021). Young’s modulus, bulk modulus, shear modulus, and longitudinal modulus all experienced notable increases with rising TiO₂ content. Calculations revealed that replacing SiO₂ with TiO₂ effectively decreased photon buildup within the material.

The present work aims to design new glass samples with the chemical composition (63-x)B₂O₃+15SiO₂+10ZnO+12Li₂O + xTiO₂, where (x = 0, 3, 6, and 10 mol%) with superior optical and radiation shielding

properties. ZnO acts as a network former in the glass matrix. Therefore, it participates in the formation of the glass network structure by forming strong covalent bonds with other network formers like B₂O₃ and SiO₂. This contributes to the overall stability and rigidity of the glass. Li₂O acts as a network modifier in the glass matrix and disrupts the network formed by B₂O₃, SiO₂, and ZnO by introducing weaker ionic bonds. This led to a lower melting point and potentially improved the workability of the glass. TiO₂ was used to modify the refractive index of the glass, allowing for tailoring the optical properties of the glass for various applications like lenses or waveguides. Moreover, the high atomic number of TiO₂ led to more effective interaction with radiation through mechanisms like Compton scattering and photoelectric absorption. This makes the glass suitable for applications requiring protection from X-rays, gamma rays, and neutrons. Borosilicate glass samples with varying TiO₂ content (0, 3, 6, and 10 mol%) were prepared using a melting-quenching technique. The structural properties were characterized by X-ray diffraction (XRD) and Raman spectroscopy, and the density was measured by the Archimedes technique. The investigation of optical properties was studied using UV–Vis spectrophotometry. Finally, the Phy-X software was used to simulate how radiation interacts with the materials by calculating factors like absorption and scattering of photons across a wide range of energy levels, including both X-rays and gamma rays.

2. Experimental technique

Glass samples with the chemical composition (63-x) B₂O₃+15SiO₂+10ZnO+12Li₂O + xTiO₂, where (x = 0, 3, 6, and 10 mol %), were prepared using analytical grade chemicals of boron oxide (B₂O₃), silicon dioxide (SiO₂), zinc oxide (ZnO), lithium carbonate (Li₂CO₃), and titanium oxide (TiO₂). Through the use of the melting-quenching technique, borosilicate glass samples containing different TiO₂ concentrations have been obtained. The precise stoichiometric chemical composition was used to determine the weight of each component. The batch was then thoroughly blended and mixed for around 60 min to preserve the homogeneity of the combined components. Following 30 min of heating to 1080 °C, the resulting mixes were poured into a stainless-steel cylinder to produce glass discs. In order to minimize thermal stress, the glass samples were annealed for 4 h at 300 °C and then allowed to cool to ambient temperature.

A Bruker D8 Advance Cu_{Kα} radiation with a wavelength of 1.5418 Å was used to collect the X-ray diffraction (XRD) patterns from 10° to 80°. The Alpha 300 AR High Resolution Raman Microscope (WITec GmbH, Germany) and a 35 mW He–Ne laser were used to perform the Raman measurements. The confocal pinhole was formed with a 50 µm diameter optical cable and an excitation laser at 633 nm. The Archimedes technique is useful because it allows for the measurement of the density of irregularly shaped objects, which can be challenging to measure directly using geometric methods. Toluene is a liquid chosen as the medium for submerging the glass because it doesn’t react with the glass and has a known density ($\rho_t = 0.866 \text{ cm}^3/\text{g}$), which is needed for the density calculation. By measuring the weight of the glass in air (W_a) and then submerged in toluene (W_t), we can determine the density of the glass (ρ) (Taha et al., 2024a).

$$\rho = \rho_t \frac{W_a}{W_a - W_t} \quad (1)$$

The ultraviolet–visible absorption spectra of borosilicate glasses doped with TiO₂ were examined using a Cary 5000 UV–Vis–NIR double beam spectrophotometer operating at room temperature. The glass samples were prepared in a suitable form (thin polished discs) for spectroscopic analysis. Background absorption was corrected using an air blank as a reference sample. Phy-X software was specifically designed to simulate the behavior of radiation interacting with matter (Şakar et al., 2020; Kaky et al., 2023; Sayyed et al., 2023a; Kaky and Sayyed, 2024a; Sayyed and Kaky, 2023). It might calculate factors like

the absorption and scattering of photons by different shielding materials across a range of energy levels. This covers a wide range of photon energies. Low-energy photons (around 0.015 MeV) are often referred to as X-rays, while higher energy photons (above 1 MeV) are categorized as gamma rays. By simulating across this range, we have assessed the performance of our present glass samples against a broad spectrum of radiation threats.

3. Results and discussion

3.1. X-ray diffraction

Fig. 1 presents the X-ray diffraction (XRD) patterns of TiO₂-doped borosilicate glasses. The absence of distinct crystalline peaks in the XRD pattern confirms the amorphous nature of the glass samples. The patterns also show diffuse scattering humps at $2\theta = 25^\circ$ – 30° and 40° – 50° , which is indicative of borate glass. As TiO₂ is incorporated, the titanium ions might occupy positions within the glass network that were previously occupied by other elements (like boron or silicon). This led to a distortion of the local atomic arrangements and a decrease in the average ordering.

The addition of TiO₂ to the borosilicate glass samples leads to a broadening of the broad humps in the XRD spectra, suggesting a disruption of the short-range order within the glass network as the TiO₂ content increases.

3.2. Raman spectroscopy

The most important technique for determining bond composition and defects in crystal symmetry is Raman spectroscopy. It is employed to examine the glass sample's vibrations, chemical composition, and structural characteristics. The Raman spectra of borosilicate glasses doped with TiO₂ in the spectral range of 100–2000 cm⁻¹ at room temperature can be observed in Fig. 2. A Boson peak below 200 cm⁻¹ is visible in the spectra of the glasses; the Boson peak above the Debye level indicates the presence of an extra vibrational density of states (Schmid and Schirmacher, 2008; Duval et al., 2007). The broad Raman peaks confirm the amorphous nature of the glass samples by indicating the considerable disorder of the glasses under examination. The collective modes of local structures or vibration modes of heavy metals are responsible for the low frequency region (≤ 250 nm) of Raman spectra, whereas the stretching vibration modes of glass network formers compose the high frequency region (Desirena et al., 2009). The ring angle bending (B–O–B) from borate units is responsible for the band at

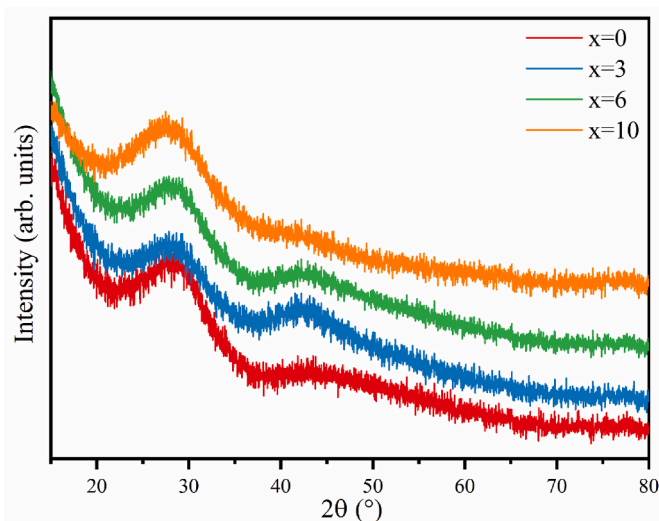


Fig. 1. X-ray diffraction patterns for borosilicate glasses doped with TiO₂.

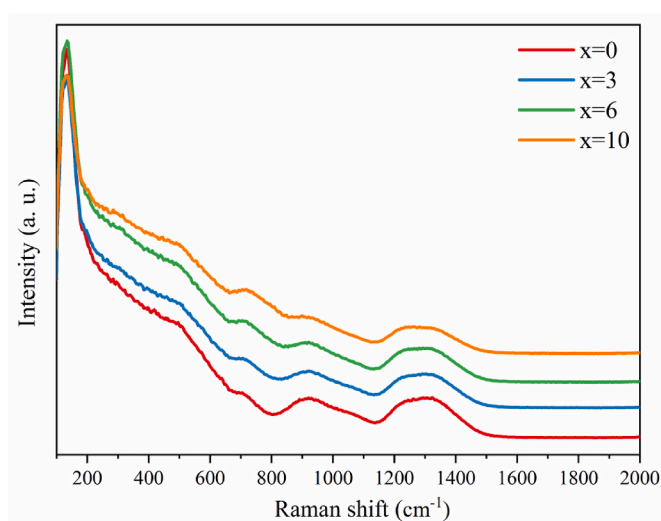


Fig. 2. Raman spectra of borosilicate glasses doped with TiO₂ in the spectral range of 100–2000 cm⁻¹.

498 cm⁻¹ in the low-frequency region less than 600 cm⁻¹ (Zhou et al., 2008). Metaborates with non-bridging oxygen groups of the chain and ring types are responsible for the bands at 720 cm⁻¹. TiO₂ acts as a network modifier, disrupting the boron-oxygen network by forming TiO₆ octahedra, which increases the number of non-bridging oxygens and alters the glass structure. This structural modification introduces new vibrational modes that interact with the existing B–O–B vibrations, shifting the 720 cm⁻¹ band to higher wavenumbers. The B–O stretching vibration of tetrahedral BO₄ in B₃O₃³⁻ (orthoborate) units correspond to the band at 930 cm⁻¹ (Padmaja and Kistaiah, 2009). The Si–O–Si bond vibrations in the glass network are most likely the cause of the Raman peaks in the calcium lithium silicate borate glass that were seen at 990 and 1100 cm⁻¹ (Osipov et al., 2013). The Raman spectra presented in Fig. 2 demonstrated that the Raman intensity decreased with the addition of Ti ions to the glass host matrix. One possible explanation for the observed decrease in Raman intensity with an increasing TiO₂ substitution ratio is a corresponding decrease in the concentration of tri-coordinated boron species inside the glass matrix (Wan et al., 2008). This clearly indicates that the structural changes are the result of the addition of TiO₂ to the borosilicate glasses.

The observed decrease in Raman intensity with increasing Ti content suggests a weakening of the vibrational bonds within the glass network due to the disruptive effect of Ti ions on the local atomic arrangements and potentially their higher polarizability compared to the replaced elements.

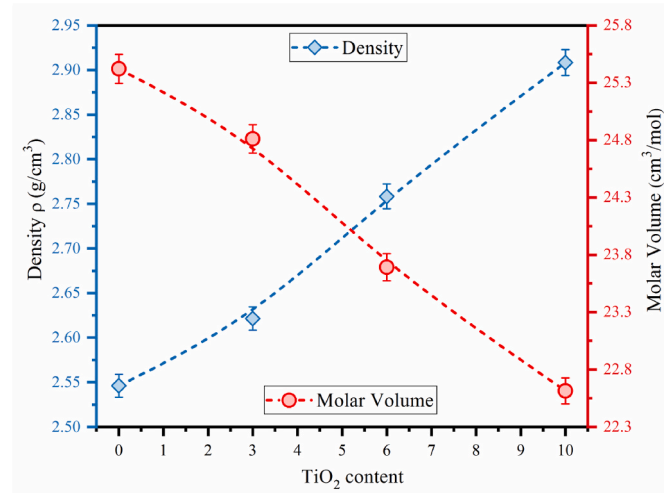
3.3. Density and molar volume

Measurements of density and molar volume are necessary for evaluating the geometric configurations of the glass network with respect to variations in its structural compactness. According to Table 1 and Fig. 3, the bulk density of the produced borosilicate glass samples rises as the concentration of TiO₂ increases, while the molar volume decreases. The improvement in density values is caused by the replacement of B₂O₃, which has a lower molecular weight and density (M.W = 69.62 g/mol – $\rho = 2.46$ g/cm³), with TiO₂, which has a higher molecular weight (M.W = 79.87 g/mol – $\rho = 4.23$ g/cm³). In addition, the increased density and compaction of the glass network's structure are due to the conversion of [BO₃] to [BO₄] groups caused by an increase in non-bridging oxygens (NBOs) inside the glass network and a decrease in boron-boron separation. Furthermore, the decrease in the molar volume demonstrated that the interatomic spacing inside the glass network had been reduced by the Ti ions. As the concentration of Ti ions increases in the current

Table 1

Chemical composition, density and molar volume of TiO₂ doped borosilicate glass samples.

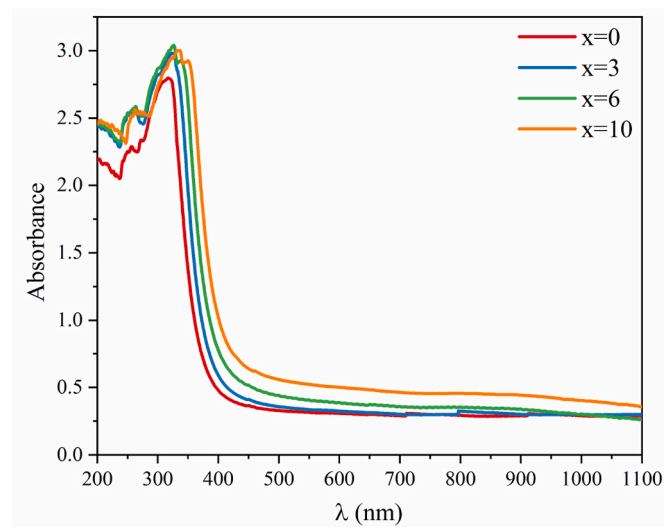
Sample	Chemical composition mol%					Molecular weight	Density (gm/cm ³)	Molar volume (cm ³ /mol)
	B ₂ O ₃	SiO ₂	ZnO	Li ₂ O	TiO ₂			
x = 0	63	15	10	12	0	64.5992	2.5462 ± 0.0001	25.3708
x = 3	60	15	10	12	3	64.9067	2.6214 ± 0.0001	24.7603
x = 6	57	15	10	12	6	65.2142	2.7584 ± 0.0001	23.6420
x = 10	53	15	10	12	10	65.6242	2.9083 ± 0.0001	22.5645


Fig. 3. Density and molar volume of borosilicate glasses doped with TiO₂.

glasses, the average distance of boron-boron separation reduces, and this decrease is also reflected in the conversion of [BO₃] to [BO₄] (Alajerami et al., 2021).

3.4. Optical properties

Fig. 4 displays the UV–Vis absorption spectra of the borosilicate glass samples that were doped with TiO₂. Based on the UV–Vis spectra, the sample's absorbance considerably increased as the substitution ratio of Ti ions increased. The structural modifications that were implemented in the glass network and modifier caused the absorption edge to shift to a higher wavelength (red shift). As the concentration of TiO₂ increased the


Fig. 4. Influence of TiO₂ doping on the UV–Vis absorption properties of borosilicate glasses.

broad near-visible band shifted towards a higher wavelength of $\lambda_{\text{cutoff}} = 416$ nm for x = 0 to $\lambda_{\text{cutoff}} = 481$ nm for x = 10. As demonstrated by the Raman spectra and physical characteristics, more (NBOs) are formed once TiO₂ is continuously added to the glass network. NBO is more polarizable than BO due to the distinct electron distribution surrounding it. Consequently, red shift of the λ_{cutoff} that has been observed is linked to the addition of TiO₂ and the growing number of NBOs (Chen et al., 2004; Guo et al., 2016).

The optical absorption and the Davis-Mott equation (2) have been used to calculate the optical energy gaps (E_g) of the borosilicate glass samples doped with Ti ions under investigation (Davis and Mott, 1970; Alhakami et al., 2024).

$$(\alpha h\nu)^n = A(h\nu - E_g) \quad (2)$$

where n is the index (2 for direct transition and 1/2 for indirect transition), A is a constant, and α denotes the absorption coefficient. Plotting $(\alpha h\nu)^2$ and $(\alpha h\nu)^{1/2}$ vs. $h\nu$ will be provided the optical band gap energies for the borosilicate glass specimens under examination, as shown in Fig. 5a, b. The observed optical band gap has been presented in Table 2; it was discovered that the direct optical energy band gap E_g^{Direct} values ranged from 3.08 eV for x = 0–2.80 eV for x = 10 in relation to the TiO₂ concentration. Furthermore, the indirect optical energy band gap

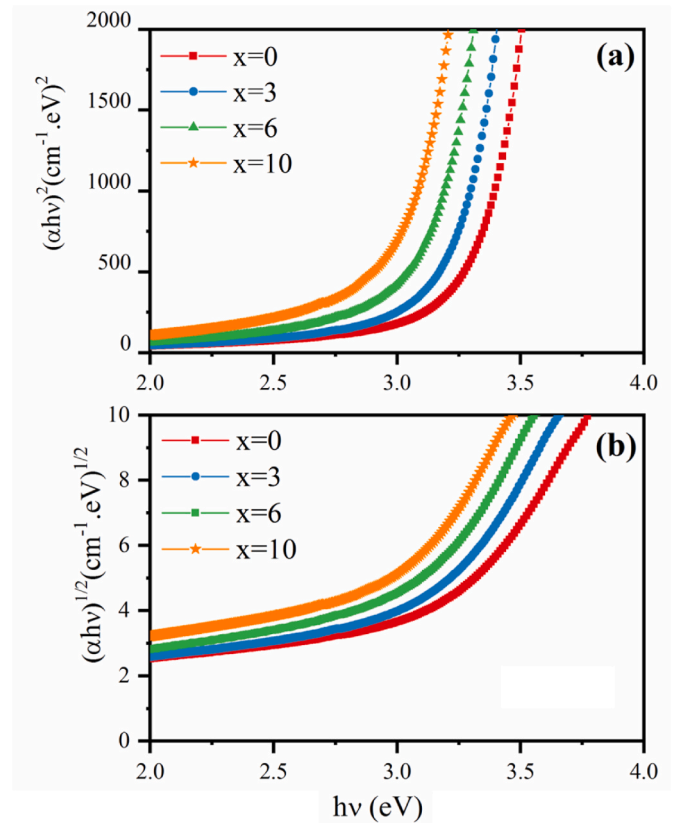

Fig. 5. Tauc plots of $(\alpha h\nu)^2$ and $(\alpha h\nu)^{1/2}$ versus photon energy ($h\nu$) for TiO₂-doped borosilicate glass samples.

Table 2Optical parameters of TiO₂ doped borosilicate glass samples.

Parameter	x = 0	x = 3	x = 6	x = 10
$\lambda_{\text{cutoff}} (\pm 0.01) \text{ (nm)}$	416	438	462	481
$E_g^{\text{Direct}} (\pm 0.01) \text{ (eV)}$	3.08	3.01	2.92	2.80
$E_g^{\text{Indirect}} (\pm 0.01) \text{ (eV)}$	2.91	2.84	2.77	2.70
$E_U (\pm 0.01) \text{ (eV)}$	0.39	0.38	0.37	0.36
$n (\pm 0.01) (\lambda = 500 \text{ nm})$	2.54	2.67	3.01	3.54
$E_d (\pm 0.01) \text{ (eV)}$	10.91	11.20	11.99	14.39
$E_0 (\pm 0.01) \text{ (eV)}$	3.82	3.54	3.48	3.31
$n_0 (\pm 0.01)$	1.96	2.04	2.11	2.31
$S_0 \cdot 10^{-5} (\pm 0.01)$	2.23	2.30	2.50	3.05
$\lambda_0 (\pm 0.01) \text{ (nm)}$	339	355	367	378

E_g^{Indirect} has been found to vary between 2.91 eV for x = 0 and 2.70 eV for x = 10.

In amorphous materials, one common tool for characterizing local states of disorder is the Urbach energy (E_u). The lower photon energy region that causes the edge absorption is often characterized using Urbach's relationship. In order to assess the Urbach energy (E_u), the band tail widths were represented by equation (3) (Almutairi et al., 2024; Urbach, 1953);

$$\ln \alpha = \ln \alpha_0 + \left(\frac{h\nu}{E_u} \right) \quad (3)$$

where α_0 is constant, and E_u is the Urbach energy. The values of the Urbach energy (E_u) are presented in Table 2 and were determined by computing the reciprocals of the slopes of the linear parts of the curves, as shown in Fig. 6. The borosilicate glasses have an Urbach energy (E_u) that varies from 0.39 eV for x = 0–0.36 eV for x = 10. The decline in E_u findings showed that the degree of disorder in the glasses reduced as the TiO₂ content increased.

Two essential optical characteristics used in the construction of optical devices are the refractive index (n) and the extinction coefficient (k). The following formula (4) was used to find the extinction coefficient (k) (Karipcin et al., 2007).

$$k = \frac{\alpha \lambda}{4\pi} \quad (4)$$

Fig. 7a shows how the extinction coefficient (k) of the produced glass samples changes as the wavelength (λ) of light varies. The graph reveals that increasing the TiO₂ concentration leads to higher k values. This

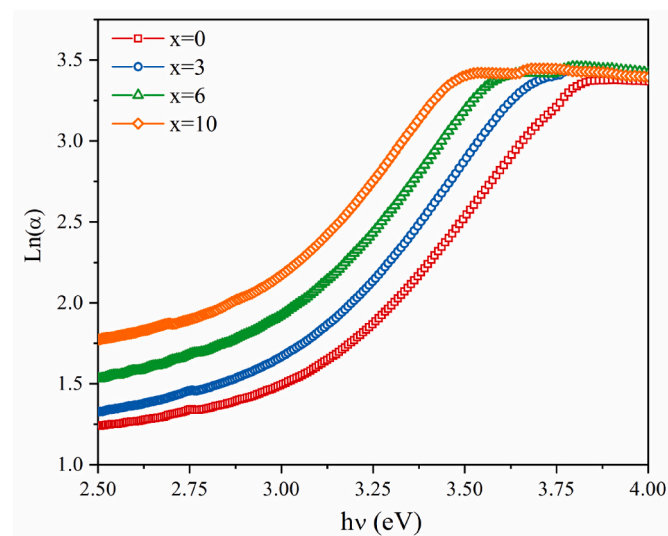


Fig. 6. Plot of $\ln(\alpha)$ versus photon energy ($h\nu$) for TiO₂-doped borosilicate glass samples.

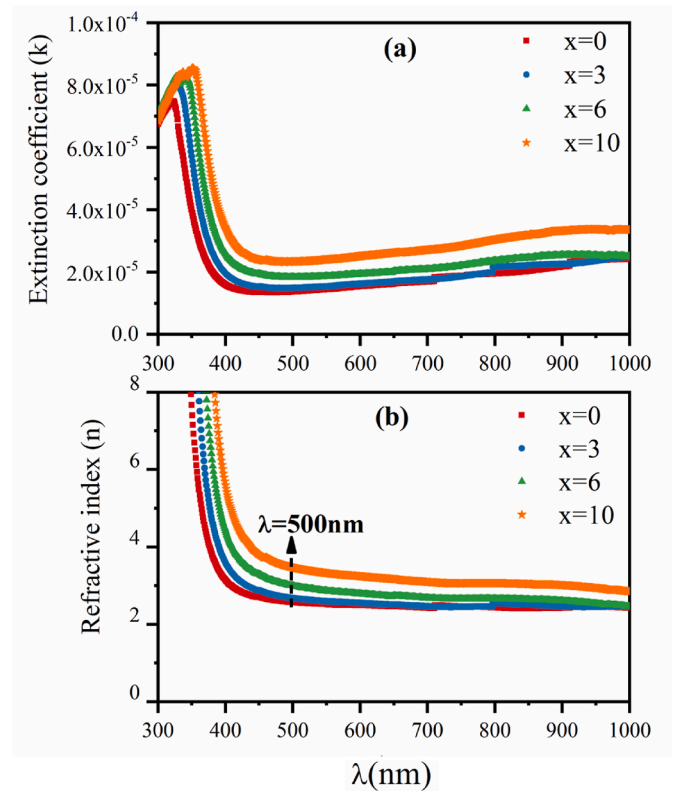


Fig. 7. Wavelength-dependent extinction coefficient (k) and refractive index (n) of TiO₂-doped borosilicate glasses.

means the glass becomes more absorbent to light. The increased absorption makes the glass suitable for applications like solar energy harvesting or optical filters.

The refractive index of borosilicate glass samples containing TiO₂ was calculated using equation (5) (El-Shamy et al., 2024).

$$n = \left(\frac{1+R}{1-R} \right) + \left(\frac{4R}{(1-R)^2} - K^2 \right)^{1/2} \quad (5)$$

Fig. 7b illustrates the refractive index of the glass samples as a function of wavelength (λ). As the TiO₂ concentration increased, so did the refractive index. This enhancement is potentially due to the formation of non-bridging oxygen within the glass network.

The dispersion energy has been explored in connection with optical materials, as it is important for optical communication and the manufacturing of devices for spectrum dispersion. The single oscillator concept was used to explore the refractive index dispersion in the glasses, which could be demonstrated using the Wemple and DiDomenico relationship (6) (Wemple et al., 1971; Taha et al., 2024b).

$$(n^2 - 1)^{-1} = \frac{E_0}{E_d} - \left(\frac{1}{E_0 E_d} \right) (h\nu)^2 \quad (6)$$

where E_0 is the average excitation energy for electronic transitions and E_d is the dispersion energy. The Wemple-DiDomenico model has been used in accordance with the experimental data demonstrated in Fig. 8. The slope of the linear fits to the experimental data from a plot of $(h\nu)^2$ dependency of $(n^2 - 1)^{-1}$ may be used to calculate E_0 and E_d , as demonstrated in Fig. 8. Table 2 lists the E_0 and E_d values for samples of borosilicate glass doped with TiO₂. The energy of the single oscillator (E_0) decreased from 3.82 eV for x = 0 sample to 3.31 eV for the x = 10 sample.

The dispersion refractive index (n_0) at the zero-photon energy point, was derived from E_0 and E_d values by applying equation (7) (Abouhaswa

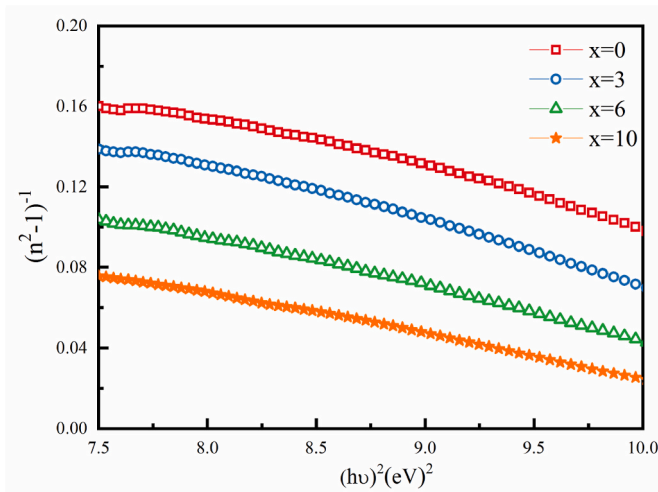


Fig. 8. Analysis of the $(n^2-1)^{-1}$ versus $(h\nu)^2$ relationship for the TiO_2 -doped borosilicate glass samples.

and Taha, 2024);

$$n_0^2 = \left(1 + \frac{E_d}{E_0}\right) \quad (7)$$

As detailed in Table 2, the estimated refractive index (n_0) exhibited a growth trend, raising from 1.96 to 2.31 as the TiO_2 substitution ratio was raised from 0 to 10 mol%.

Equation (7) was used in conjunction with the Sellmeier oscillator formula to calculate the average oscillator strength (s_0) and average inter-band oscillator wavelength (λ_0) (Parida et al., 2022);

$$(n^2 - 1)^{-1} = \left(\frac{1}{S_0 \lambda_0^2}\right) - \left(\frac{1}{S_0}\right) \lambda^{-2} \quad (8)$$

The relationship between refractive index (n) and wavelength (λ) can be visualized by plotting $(n^2-1)^{-1}$ against λ^{-2} . This method is illustrated in Fig. 9. Table 2 shows that as TiO_2 concentration increases from 0 to 10 mol%, the calculated values for the dispersion coefficients S_0 and λ_0 rise from 2.234×10^{-5} to 3.053×10^{-5} and from 339 to 378 nm, respectively.

The addition of TiO_2 significantly impacts the average oscillator strength and average inter-band oscillator wavelength of the glass samples. The increase in S_0 with TiO_2 suggests that the presence of

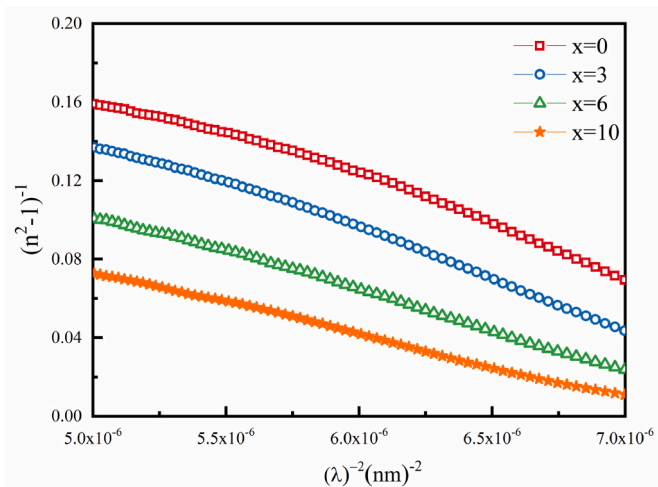


Fig. 9. The relationship between $(n^2-1)^{-1}$ and $(\lambda)^{-2}$ for the TiO_2 -doped borosilicate glass samples.

titanium ions introduces new or enhances existing electronic transitions within the glass. This could be due to the electronic structure of titanium or the way it interacts with other elements in the glass network. Moreover, the increase in λ_0 with TiO_2 addition indicates a shift in the absorption edge of the glass towards longer wavelengths. This implies that the glass with a higher TiO_2 content might absorb less ultraviolet (UV) light and potentially transmit more visible light compared to the glass without TiO_2 .

3.5. Radiation shielding

Studying the radiation shielding performance of the present TiO_2 -containing glasses is crucial for developing new and improved materials that can effectively protect people and the environment from harmful radiation exposure in various critical fields. The mass attenuation coefficient (MAC) represents the ability of a material to attenuate radiation per unit mass (Elsafi et al., 2021; Kaky and Sayyed, 2024b; Sayyed et al., 2024b). When TiO_2 is added to the glass samples, the MAC is observed to increase, as shown in Fig. 10a. Titanium (Ti) has a higher atomic number (Z) compared to the elements present in the base glass composition (B, Si, Zn, Li, and O). Elements with higher Z interact more effectively with radiation through photoelectric absorption and Compton scattering, leading to increased attenuation. Moreover, TiO_2 is denser than most of the other components in the base glass. As the TiO_2 content increases, the overall density of the glass sample will also increase. A denser material offers more target nuclei for radiation interaction, resulting in a higher MAC (Polatoğlu and Şakar, 2024).

The linear attenuation coefficient (LAC) signifies the exponential decrease in intensity of radiation as it travels through a material with a unit thickness. The LAC is related to the MAC by the equation ($\text{LAC} = \text{MAC} \cdot \rho$) (Alrowaili et al., 2021a). Therefore, the trends observed in MAC will also be reflected in LAC. As the TiO_2 content increases, the LAC is expected to increase due to the combined effect of higher Z and increased density. TiO_2 led to an increase in both MAC and LAC across the X-ray energy range. This signifies a stronger ability of the glass containing TiO_2 to attenuate X-rays compared to the base glass ($x = 0$ mol%) (Alrowaili et al., 2021b). Several research studies have investigated the effect of TiO_2 addition on the attenuation properties of glasses. These studies report an increase in both MAC and LAC with increasing TiO_2 content. For instance, a study by N. Almousa et al. (2024b) observed a significant enhancement in the gamma-ray attenuation coefficient of magnesium barium sodium borate glasses with the incorporation of TiO_2 .

Fig. 11a–d illustrates the dependence of the half-value layer (HVL), tenth-value layer (TVL), mean free path (MFP), and effective atomic number (Z_{eff}) on the radiation energy for the borosilicate glass samples doped with TiO_2 . The inclusion of TiO_2 improved the radiation shielding efficiency of the glass samples. This is evident from the decrease in HVL, TVL, and MFP and the increase in Z_{eff} . HVL represents the thickness of the material required to reduce the intensity of radiation by half (Sayyed et al., 2023b). With increasing TiO_2 content, the higher atomic number (Z) of titanium leads to stronger interactions with radiation. This translates to a decrease in HVL, as shown in Fig. 11a. In simpler terms, less thickness of the glass containing TiO_2 will be needed to achieve the same level of radiation attenuation compared to the base glass ($x = 0$ mol%). On the other hand, TVL signifies the thickness needed to reduce radiation intensity by a factor of 10. Similar to HVL, the increased Z of Ti due to TiO_2 addition led to a decrease in TVL (see Fig. 11b). This indicates a more efficient attenuation with less material required for significant radiation reduction (Abualsayed, 2023).

MFP represents the average distance a radiation particle travels before interacting with the material. As TiO_2 increases the material's density and Z, the interaction probability with radiation also rises. This results in a decrease in MFP, as shown in Fig. 11c. A shorter MFP implies more frequent interactions and stronger attenuation within the glass (Al-Burihi et al., 2021; Sayyed et al., 2024c). Finally, Z_{eff} is a weighted

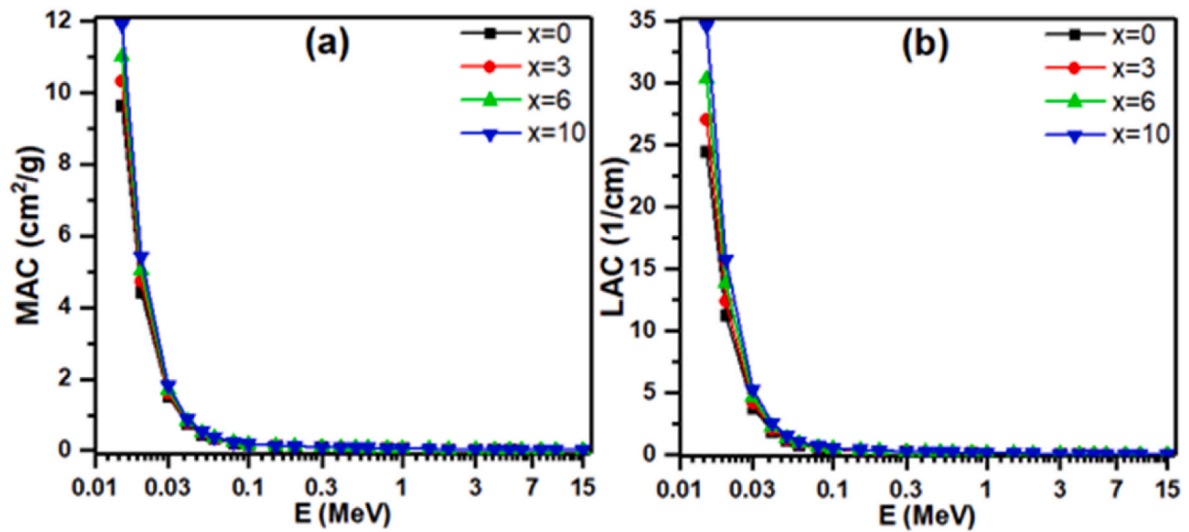


Fig. 10. The dependence of (a) MAC and (b) LAC on the radiation energy for the borosilicate glass samples doped with TiO_2 .

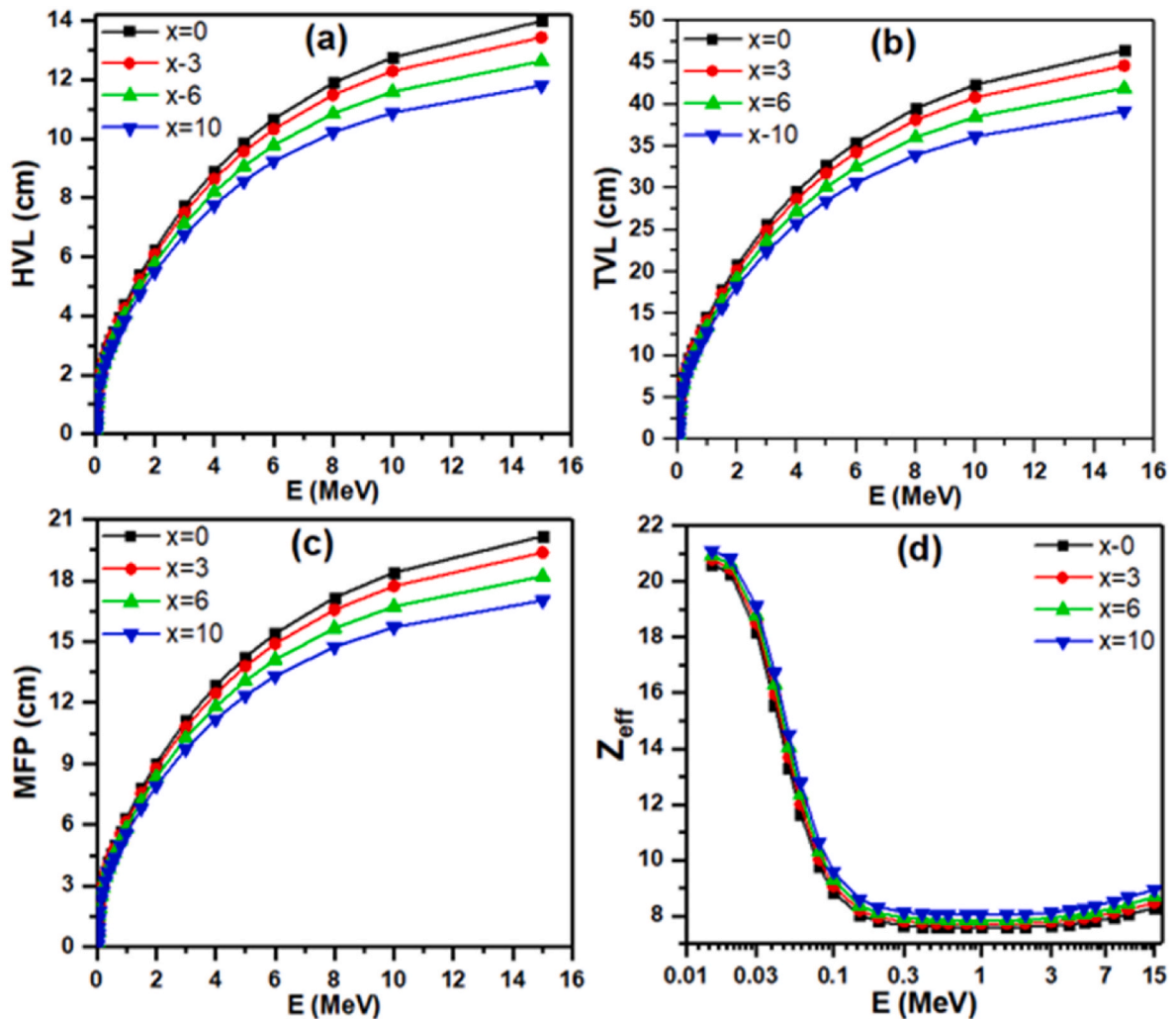


Fig. 11. The dependence of (a) HVL, (b) TVL, (c) MFP, and (d) Z_{eff} on the radiation energy for the borosilicate glass samples doped with TiO_2 .

average atomic number that reflects the overall interaction capability of the material with radiation. Due to the higher Z of Ti compared to other elements in the base glass, adding TiO_2 will lead to an increase in Z_{eff}

(see Fig. 11d). A higher Z_{eff} indicates a stronger ability to attenuate radiation.

The dependence of effective electron density (N_{eff}), effective

conductivity (C_{eff}), atomic cross-section (ACS), and electronic cross-section (ECS) on the radiation energy for the borosilicate glass samples doped with TiO_2 is given in Fig. 12a–d. As TiO_2 contains elements with more electrons per atom compared to some elements in the base glass, adding TiO_2 led to a slight increase in N_{eff} (see Fig. 12a). More electrons per unit volume can lead to more interaction points for photons, influencing attenuation. There are various mechanisms by which materials shield radiation, including absorption, scattering, and reflection. The C_{eff} likely relates to the absorption mechanism, where the free electrons interact with the radiation. In the context of radiation shielding for glass, C_{eff} reflects the number of free electrons generated when the glass interacts with radiation. Fig. 12b indicates the increase in effective conductivity upon increasing the content of TiO_2 (Ghozza et al., 2024).

While the average ACS and ECS of the material changed slightly due to the presence of Ti with a different ACS and ECS than other elements, the overall effect on the bulk material's shielding might be minor (see Fig. 12c and d). This is because ACS and ECS are inherent properties of an atom and don't significantly depend on its surrounding environment in the glass network (Sabry et al., 2022).

Fig. 13 displays the variation of energy (EBF) and energy absorption (EABF) buildup factors as a function of penetration depth up to 40 MFP for the borosilicate glass samples doped with TiO_2 .

EBF represents the rate at which the intensity of radiation increases within a material as it travels through (Aloraini et al., 2023). As discussed earlier, Ti has a higher Z compared to most elements in the base

glass. This leads to more frequent interaction with radiation through Compton scattering. During Compton scattering, a portion of the radiation's energy is transferred to electrons, causing them to be ejected from their orbitals. These energetic electrons can further interact with other atoms, depositing their energy within the material (Sayyed and Elhouichet, 2017). On the other hand, EABF signifies the fraction of total radiation energy absorbed by the material. Similar to EBF, the inclusion of TiO_2 led to an increase in EABF. The increased Compton scattering due to higher Z translates to a greater conversion of radiation energy into heat within the glass. This higher conversion efficiency results in a larger fraction of the total radiation energy being absorbed by the material (Nagaraja et al., 2023). The radiation shielding performance of the $(63-x)\text{B}_2\text{O}_3 + 15\text{SiO}_2 + 10\text{ZnO} + 12\text{Li}_2\text{O} + x\text{TiO}_2$ (where $x = 0, 3, 6$, and 10 mol%) is competitive to other materials based on several key parameters. Previous studies indicated that borosilicate glasses with varying ZnO concentrations exhibit significant gamma-ray shielding capabilities, with linear attenuation coefficients (LAC) increasing with higher ZnO content, outperforming conventional materials like concrete (Saleh et al., 2024). Similarly, borate glasses incorporating dysprosium (III) oxide showed enhanced shielding properties, particularly with increased Dy_2O_3 concentrations, which improved the effective atomic number and reduced the half-value layer (HVL) (Almoussa et al., 2024c). Additionally, Bi_2O_3 -based glasses demonstrated superior mass attenuation coefficients, significantly exceeding those of commercial shielding glasses (Shah et al., 2024). In contrast, Al_2O_3 - PbO - B_2O_3 - SiO_2 - BaO glasses also showed improved shielding with increased BaO content

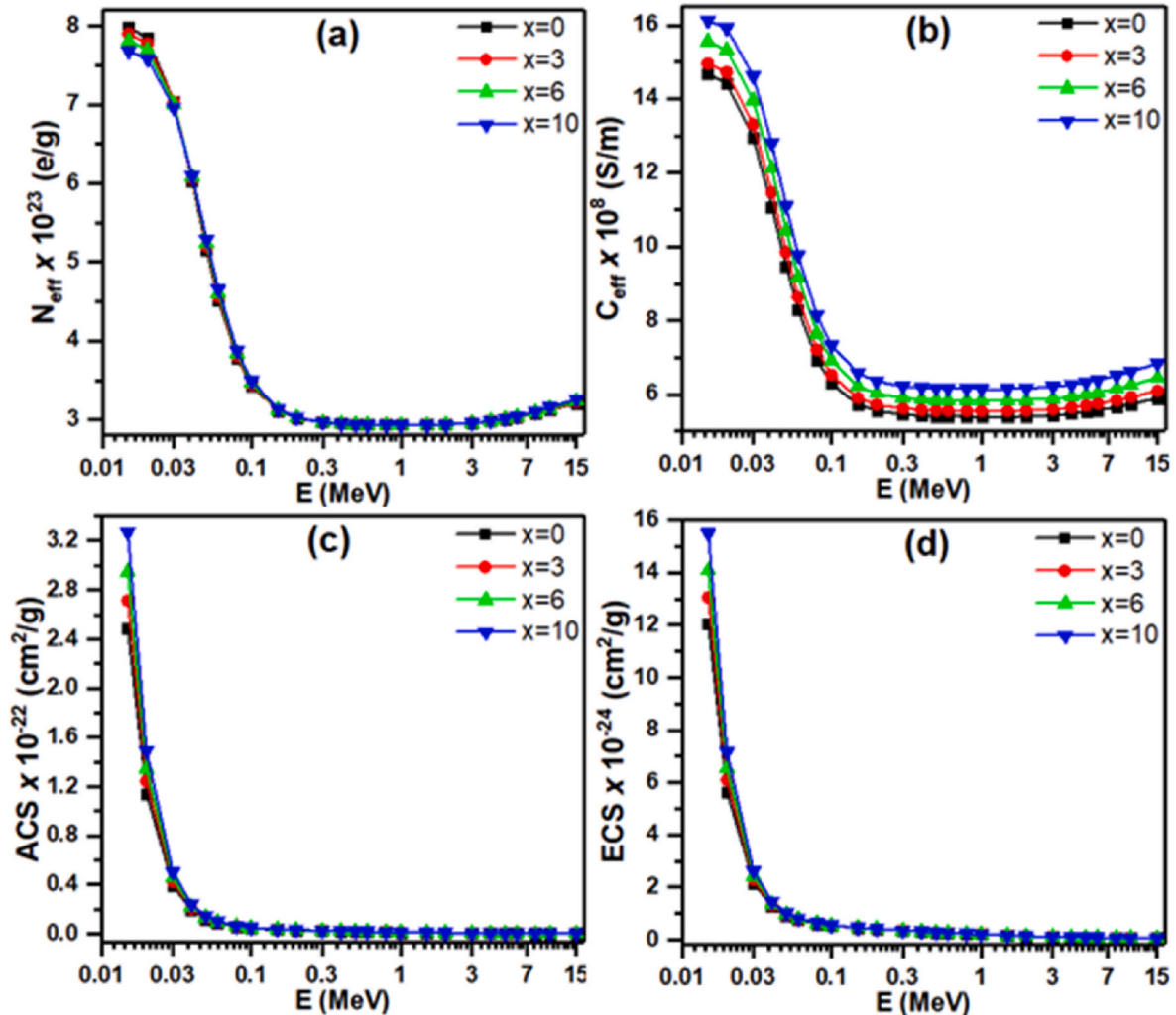


Fig. 12. The dependence of (a) N_{eff} , (b) C_{eff} , (c) ACS, and (d) ECS on the radiation energy for the borosilicate glass samples doped with TiO_2 .

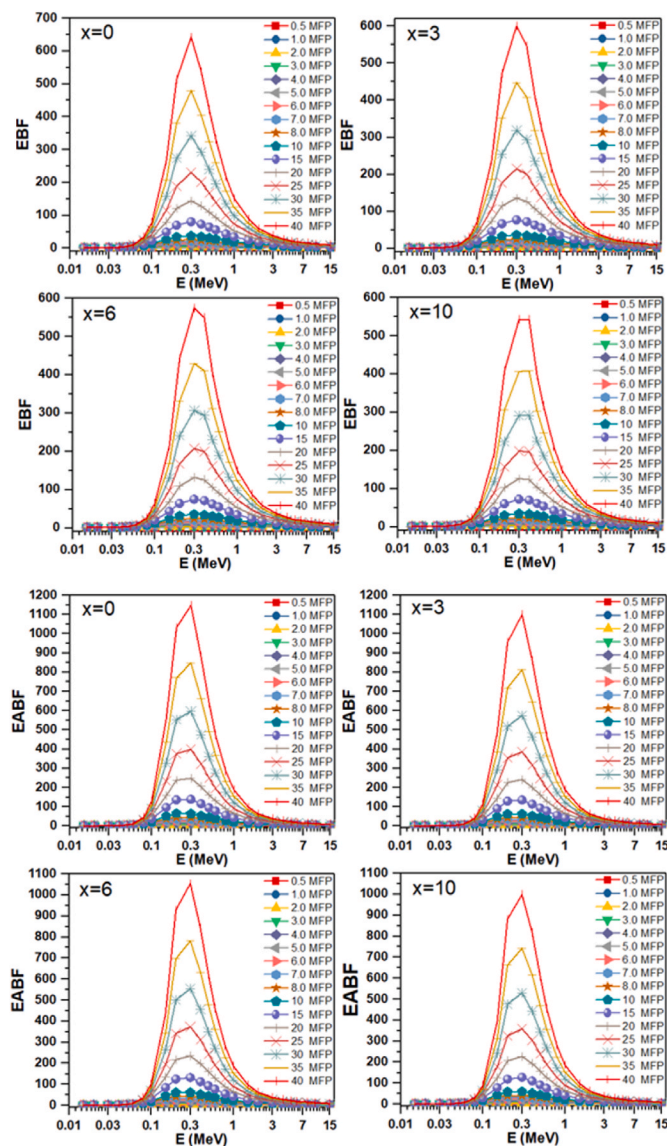


Fig. 13. The variation of EBF and EABF as a function of penetration depth up to 40 MFP for the borosilicate glass samples doped with TiO_2 .

(Almuqrin et al., 2024). Accordingly, by optimizing the TiO_2 content and considering potential trade-offs, researchers can develop glasses with tailored shielding properties for various applications.

4. Conclusions

The current study successfully designed new glass samples with the chemical composition $(63-x)\text{B}_2\text{O}_3 + 15\text{SiO}_2 + 10\text{ZnO} + 12\text{Li}_2\text{O} + x\text{TiO}_2$ (where $x = 0, 3, 6$, and 10 mol%) with superior optical and radiation shielding properties. XRD and Raman spectroscopy confirmed the amorphous nature of the glass samples. The addition of TiO_2 to the borosilicate glass matrix increased the density (2.5462 – 2.9083 g/cm³). The presence of TiO_2 led to a red shift in the UV–Vis spectra and a decrease in the indirect optical band gap (2.91 – 2.70 eV). The Urbach energy values decreased (0.39 – 0.36 eV) with increasing TiO_2 concentration. Refractive index (n_0) increased monotonically from 1.96 to 2.31 with increasing TiO_2 content. A systematic decrease in the single oscillator energy (E_0) from 3.82 eV to 3.31 eV was observed as the TiO_2 content increased from $x = 0$ to $x = 10$. The dispersion coefficients S_0 and λ_0 rise from 2.234×10^{-5} to 3.053×10^{-5} and from 339 to 378 nm, respectively. HVL, TVL, and MFP increased with TiO_2 content,

indicating less material thickness is needed for the same level of attenuation. Z_{eff} was improved and signified a stronger ability to attenuate radiation. EBF and EABF were enhanced after the addition of TiO_2 , indicating a more efficient deposition of radiation energy within the glass, making it a better radiation shield. The addition of TiO_2 presented a promising approach to enhancing both the optical and radiation shielding performance of these glasses.

Author's statement

All the authors have contributed to the work and are known with the submission.

CRediT authorship contribution statement

Mohammad H. Alhakami: Project administration, Investigation, Funding acquisition, Conceptualization. **A.S. Abouhaswa:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Numa A. Althubiti:** Validation, Software, Project administration, Investigation. **Taha Abdel Mohaymen Taha:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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